MILITARY SPECIFICATION

MOUNT, TELESCOPE: 11741796

"This specification is approved for use by all departments and agencies of the Department of Defense".

1. SCOPE

1.1 This specification covers one type of mount known as Mount, Telescope: 11741796 used with the Lightweight Company Mortar 60MM, XM224 (See 6.1). Throughout this specification this instrument is referenced as "mount". This mount has radio-actively illuminated scales and level vials.

2. APPLICABLE DOCUMENTS

2.1 The following documents, of the issue in effect on date of invitation for bids or request for proposals, form a part of this specification to the extent specified herein.

SPECIFICATIONS

Military

MIL-F-13926	Fire Control Materiel: General Specification Governing the Manufacture and Inspection of
MIL-P-14232	Parts, Equipment and Tools for Army Materiel, Packaging and Packing Of
MIL-I-45607	Inspection Equipment, Supply and Maintenance Of
MIL-L-50778	Liquid Scintillation System for Radioactive Fire Control

STANDARDS

Military

MIL-STD-105	Sampling Procedures and Tables for Inspection by Attributes
MIL-STD-109	Quality Assurance Terms and Definitions
MIL-STD-810	Environmental Test Methods
MIL-STD-1458	Radioactive Materials: Marking and Labeling of Items, Packages and Shipping Containers for Identification in Use Storage and Transportation

Beneficial comments (recommendations, additions, deletions) and any pertinent data which may be of use in improving this document should be addressed to: Commander, ARRADCOM, ATTN: DRDAR-TST-S, Dover, N.J. 07801 by using the self-addressed Standardization Document Improvement Proposal (DD Form 1426) appearing at the end of this document or by letter.

Source: https://assist.dla.mil -- Downloaded: 2026-10-26T03:56Z
Check the source to verify that this is the current version before use.

DRAWING

US Army, Frankford Arsenal

F11741796	Mount, Telescope: 11741796
F6523553	Elevation and Cross Leveling Fixture
F7680631	Dioptometer
F7691596	Azimuth Test Fixture
F11750708	Adapter, Final Inspection, Elevation
F11750709	Adapter, Final Inspection, Azimuth
F11750712	Alignment Gage, Final Inspection, Azimuth

PACKAGING DATA SHEETS

11741796	Packaging of Mount Telescope: 11741796

(Copies of specifications, standards, drawings and packaging data sheets required by suppliers in connection with specific procurement functions should be obtained from the procuring activity or as directed by the contracting officer.)

2.2 Other publications.- The following documents form a part of this specification to the extent specified herein. Unless otherwise indicated the issue in effect on date of invitation for bids or request for proposal shall apply.

DEPARTMENT OF TRANSPORTATION, BUREAU OF HAZARDOUS MATERIAL

49 CFR 171-190 - Interstate Commerce Commission and Department of Transportation Rules and Regulations for the Transportation of Explosives and other Dangerous Articles.

(Application for copies should be addressed to the Superintendent of Documents, Government Printing Office, Washington, D.C. 20402.)

CODE OF FEDERAL REGULATIONS

Title 10 (Nuclear Regulatory)
Part 20: Standards for protection against radiation.
Part 30: Rules of general applicability to licensing of by-product material.

(Application for copies should be addressed to the Superintendent of Documents, Government Printing Office, Washington, D.C. 20402.)

3. REQUIREMENTS

3.1 Radioactive material.- Prospective bidders should be cognizant of the need for a Nuclear Regulatory Commission License governing the procurement possession, handling and shipping of radioactive material.

Source: https://assist.dla.mil -- Downloaded: 2026-10-26T03:56Z
Check the source to verify that this is the current version before use.

3.1.1 Regulations pertaining to radioactive devices.- Marking, identification, handling, and shipping of radioactive devices shall be in accordance with MIL-STD-1458.

3.1.2 External tritium contamination.- Any part of tritium shall not be present on the external surfaces of the mount in excess of 1000 dpm/100 sq. cm. subsequent to each of these requirements: 3.7.1, 3.7.2, 3.7.3 and 3.7.4.

3.1.3 Illumination check.- Illuminated areas shall be given a visual check to determine if the light sources are illuminating subsequent to each of these requirements: 3.7.1, 3.7.2, 3.7.3 and 3.7.4.

3.2 Materials.- Materials shall be in accordance with drawings, material specifications and general specifications forming a part of this specification.

3.3 Fabrication.- The mount shall be manufactured in accordance with Drawing F11741796.

3.4 General specifications.- The following requirements of MIL-F-13926 shall be met:

a. Order of Precedence
b. Dimensions and Tolerances (See USASI Y14.5)
c. Inorganic protective surface finishes
d. Parts Identification and Marking
e. Workmanship
f. Bonded Joints

3.5 Temperature conditions.- The requirements of this specification shall be met with the mount thermally stabilized at an ambient temperature of between 60° and 90° Fahrenheit (F) unless otherwise specified in the requirement paragraphs.

3.6 Workmanship.- The workmanship requirements of MIL-F-13926 shall be met. The exterior protective surface finishes shall be free of any defects which effect their function protective qualities or appearances.

3.7 Environmental.- Any mount subjected to any or all of the requirements of 3.7.1, 3.7.2, 3.7.3 and 3.7.4 inclusive, shall subsequently meet the requirements of 3.9 to 3.17 inclusive, unless otherwise specified.

3.7.1 Storage temperature.- The mount shall show no evidence of physical failure such as potting separations, glass vial breakage or casting failure after having been exposed and thermally stabilized at 50° and +160°F.

3.7.2 Operating extremes.- The mount shall meet the requirements of 3.16 to 3.16.5 inclusive, while exposed to and thermally stabilized at ambient temperatures of -50°F and +145°F.

Source: https://assist.dla.mil -- Downloaded: 2026-10-26T03:56Z
Check the source to verify that this is the current version before use.

3.7.3 Shock.- The mount shall be shocked six (6) times in each of three positions. The elevation scale shall be set at the 800 mil mark and the locking knobs turned to a locked position with a ten pound-inch torque. The applied shocks shall be in a vertical, up direction. In the first position the mounting dovetail shall be in the horizontal plane and subjected to six (6) shocks, each at 300g for one-half millisecond. In the second and third positions, the mounting dovetail shall be in a vertical plane with its axis at 45° and 225° and subjected to six (6) shocks each at 200g for one millisecond. The shock pulse shall be a one-half sine wave.

3.7.4 Vibration.- The mount shall be vibrated at 30 ± 5 cycles per second frequency with an amplitude of 1/16 inch (1/8 inch maximum excursion) for a period of five minutes ± 15 seconds. Orientation of the mount shall be in accordance with requirement 3.8. In addition, the locking knob should be turned to a locked position with a 10 pound-inch torque.

3.8 Orientation.- The mount shall be positioned as follows to meet the requirements of 3.9 to 3.16.5 inclusive. Where line of sight is specified, the requirement shall be met with the line of sight entering the telescope mounting sleeve from the end opposite to the telescope elevation mechanism.

(a) The mounting dovetail surface in a vertical plane.
(b) The axis of the dovetail vertical.
(c) A line of sight provided which is parallel to the above mounting surface and which is perpendicular to the above axis.
(d) The elevation and cross level vial bubbles shall be centered within a half of one graduation space.
(e) The coarse azimuth scale index indicating 3200 mils within a half of the width of the graduation line.
(f) The coarse elevation scale index indicating 800 mils within a half of the width of the graduation line.
(g) The indices for the graduated scales on the azimuth and elevation knobs indicating zero within 0.5 mil.

3.9 Elevation travel.- Rotation of the elevation knob shall cause the line of sight to depress a minimum of 800 mils and elevate a minimum of 100 mils from the position specified in 3.8.

3.10 Plumb travel.- The line of sight shall track a plumb line within 1.0 mil when elevated and depressed through the range specified in 3.9 and the position specified in 3.8. The total lateral spread shall not exceed 1.0 mil.

3.11 Lateral displacement.- The line of sight shall not exceed 0.5 mil lateral displacement when the elevation motion is reversed by means of the elevation knob.

3.12 Vertical displacement.- The line of sight shall not exceed 0.5 mil vertical displacement when the azimuth motion is reversed by means of the azimuth knob.

Source: https://assist.dla.mil -- Downloaded: 2026-10-26T03:56Z
Check the source to verify that this is the current version before use.

3.13 Horizontal travel.- The line of sight shall not vary in elevation in excess of 1.0 mil (Total spread of 2 mils) when rotated 6400 mils in azimuth by means of the azimuth knob.

3.14 Angular accuracy.

3.14.1 Elevation.- The elevation scales shall indicate elevation angles within 1.0 mil, excluding backlash at each reading, when the line of sight is elevated and depressed through the range specified in 3.9.

3.14.2 Azimuth.- The azimuth scales shall indicate azimuth deflection within 1.0 mil, excluding backlash at each reading, when the line of sight is rotated through 6400 mils in azimuth.

3.15 Backlash.

3.15.1 Elevation.- When the amount of backlash at each elevation reading, taken in accordance with 3.14.1 is added to the elevation error at that setting, the sum shall not exceed 2.0 mil.

3.15.2 Azimuth.- When the amount of backlash at each azimuth reading, taken in accordance with 3.14.2 is added to the azimuth error at that setting, the sum shall not exceed 2.0 mil.

3.16 Operability.

3.16.1 Telescope clamping mechanism.- The telescope clamping mechanism shall be operable in the unlocked and locked positions when using a mandrel test piece having a diameter of .624 - .001 inches. These requirements shall be met at the ambient temperature of 3.5 and while the mount is subjected to the operating temperatures of 3.7.2.

3.16.1.1 Unlocked position.- The mandrel test piece shall rotate freely within its mating diameter and the clamping mechanism shall operate freely about its axis in a fore and aft direction.

3.16.1.2 Locked position.- There shall be no rotation of the mandrel within its mating diameter when a 35 inch-pound torque is applied. The clamping mechanism shall not rotate about its axis when a 12 inch-pound torque is applied.

3.16.2 Mount locking latch.- The mount locking latch shall be operable at the ambient temperature of 3.5 and while subjected to the operating temperatures of 3.7.2.

3.16.3 Locking knobs.- Subsequent to rotating the elevation and azimuth locking knobs to a locked position with a 10 pound-inch torque, there shall be no movement in their respective mechanisms when an 8 pound-inch torque is applied to the elevation and azimuth knobs. This requirement shall be met at the ambient temperature of 3.5 and while the mount is subjected to the operating temperatures of 3.7.2.

Source: https://assist.dla.mil -- Downloaded: 2026-10-26T03:56Z
Check the source to verify that this is the current version before use.

3.16.4 Azimuth slip scales.- The micrometer and azimuth slip scales shall be operable at the ambient temperature of 3.5 and while subjected to the temperature requirements of 3.7.2.

3.16.5 Torque.- The running torque in inch-pounds on all knobs shall be within the limits specified in Table I.

TABLE I

TORQUE VALUES (INCH-POUNDS)

KNOB	+60° to 90°F (STANDARD TEMPERATURE)		-50° and +145° F (EXTREME OPERATING TEMPERATURE)	
	Min	Max	Min	Max
Elevation	0.5	4.0	0.5	9.0
Azimuth	0.5	4.0	0.5	9.0

3.17 Illumination.- The markings of the scales indices and level vials should be clearly distinguishable when observed in ambient light conditions ranging from dusk into night at a distance of 18 to 20 inches.

3.18 Reliability.- The mount shall have a minimum Mean Rounds Between Failure (MRBF) of 5000 rounds and based on a mission of 210 rounds, the probability of success is 96 percent (96%).

4. QUALITY ASSURANCE PROVISIONS

4.1 Responsibility for inspection.- Unless otherwise specified in the contract or purchase order, the supplier is responsible for the performance of all inspection requirements as specified herein. Except as otherwise specified, the supplier may utilize his own facilities or any other facilities suitable for the performance of the inspection requirements specified herein, unless disapproved by the Government. The Government reserves the right to perform any of the inspections set forth in the specification where such inspections are deemed necessary to assure supplies and services conform to prescribed requirements.

4.1.1 General provisions.- The component and subassembly inspection requirements of MIL-F-13926 form a part of the Quality Assurance Provisions of this specification. Definitions of inspection terms shall be as listed in MIL-STD-109.

4.2 First article (initial production) approval.- The requirement for first article approval and the responsibility (Government or contractor) for first article testing shall be as specified in the contract. The sample for first article approval tests shall consist of three (3) mounts plus three (3) each of all items covered by SQAP. The three mounts shall be tested as specified herein for all the requirements of this specification. The items covered by SQAP shall be tested, as specified thereon, for all characteristics delineated on the SQAP. The samples shall be manufactured in the same manner, using the same materials, equipment, processes,

Source: https://assist.dla.mil -- Downloaded: 2026-10-26T03:56Z
Check the source to verify that this is the current version before use.

and procedures as used in regular production. All parts and materials, including packaging and packing, shall be obtained from the same source of supply as used in regular production.

4.2.1 Government testing.- When the Government is responsible for conducting first article approval tests, the contractor prior to submitting the sample to the Government, shall inspect the sample to insure that it conforms to all the requirements of the contract and submit a record of this inspection with the sample, including certificates of conformance for materials.

4.2.2 Contractor testing.- When the contractor is responsible for conducting first article approval tests, the sample shall be inspected by the contractor for all the requirements of the contract. The sample and a record of this inspection, including certificates of conformance for materials, shall be submitted to the Government for approval. The Government reserves the right to witness the contractor's inspection.

4.3 Inspection provisions.

4.3.1 Submission of product.- Unless otherwise specified by the Contracting Officer, inspection lot size, lot formation and presentation of lots shall be in accordance with "Submission of Product" provisions of MIL-STD-105.

4.3.2 Examination and tests.

4.3.2.1 Components and subassemblies.- All components and subassemblies shall be inspected in accordance with the inspection provisions contained in Supplementary Quality Assurance Provisions (SQAP) listed in the technical data package (TDP). In the absence of SQAP's, the applicable Quality Assurance Provisions of MIL-F-13926 shall apply.

4.3.2.2 Mount.- Examination and tests related to Section 3 herein shall be performed on a class basis in accordance with MIL-STD-105 and the sampling plans specified in Tables II, III, and IV herein. Examination and tests for packaging, packing, and marking shall be in accordance with MIL-P-14232 and Section 5 herein. The tabulated classification of defects in Tables II, III, and IV shall constitute the minimum inspection to be performed by the supplier after first article approval and prior to Government acceptance or rejection by item or lot.

TABLE II - CLASSIFICATION OF DEFECTS

CLASS CRITICAL: NONE MAJOR: AQL 1.0% Defective	REQUIREMENT	TEST PROCEDURE
101. Vibration	3.7.3	4.6.4
102. External tritium contamination	3.1.2	4.6.5
103. Illumination	3.1.3	Visual

Source: https://assist.dla.mil -- Downloaded: 2026-10-26T03:56Z
Check the source to verify that this is the current version before use.

104.	Elevation travel	3.9	4.6.6.1
105.	Plumb travel	3.10	4.6.6.2
106.	Lateral displacement	3.11	4.6.6.3
107.	Elevation angular accuracy	3.14.1	4.6.6.4
108.	Elevation backlash	3.15.1	4.6.6.5
109.	Horizontal travel	3.13	4.6.7.1
110.	Vertical displacement	3.12	4.6.7.2
111.	Azimuth angular accuracy	3.14.2	4.6.7.3
112.	Azimuth backlash	3.15.2	4.6.7.4
113.	Telescope clamping mechanism, unlocked position	3.16.1.1	4.6.8.1
114.	Telescope clamping mechanism, locked position	3.16.1.2	4.6.8.2
115.	Mount locking latch	3.16.2	4.6.8.3
116.	Locking knobs	3.16.3	4.6.8.4
117.	Azimuth slip scales	3.16.4	4.6.8.5
118.	Elevation knob torque	3.16.5	4.6.8.6
119.	Azimuth knob torque	3.16.5	4.6.8.6
120.	Illumination	3.17	4.6.9
121.	Workmanship	3.6	4.6.10

MINOR: NONE

NOTE: The tests in Table II shall be conducted at a temperature between +60°F and +90°F.

4.3.3 Acceptance and rejection.- Rejected lots shall be screened for all defective characteristics. Removal or correction of defective units and resubmittance of rejected lots shall be in accordance with "Acceptance and Rejection" as specified in MIL-STD-105.

4.4 Special sampling.

4.4.1 General.- One mount shall be selected at random as a special sample from each 100 produced. The sample shall meet the requirements and tests in Table III.

TABLE III - CLASSIFICATION OF DEFECTS

NO.	CHARACTERISTIC	REQUIREMENT	TEST PROCEDURE
301.	Fabrication	3.3	Applicable drawing - visual
302.	General specification	3.4	MIL-F-13926 - visual

NOTE: The tests in Table III shall be conducted at a temperature between +60° and +90°F.

Source: https://assist.dla.mil -- Downloaded: 2026-10-26T03:56Z
Check the source to verify that this is the current version before use.

4.4.2 Environmental.- Three mounts shall be selected at random as samples from each 50 produced, or from each month's production, whichever occurs first. Each sample shall meet the requirements and tests in Table IV and shall meet the requirements and tests in Table II.

TABLE IV - CLASSIFICATION OF DEFECTS

NO.	CHARACTERISTIC	REQUIREMENT	TEST PROCEDURE
303.	Storage temperature -50°F and +160°F	3.7.1	4.6.1
304.	Telescope clamping mechanism, unlocked position	3.7.2, 3.16.1.1	4.6.2, 4.6.8.1
305.	Telescope clamping mechanism, locked position	3.7.2, 3.16.1.2	4.6.2, 4.6.8.2
306.	Mount locking latch	3.7.2, 3.16.2	4.6.2, 4.6.8.3
307.	Locking knobs	3.7.2, 3.16.3	4.6.2, 4.6.8.4
308.	Azimuth slip scales	3.7.2, 3.16.4	4.6.2, 4.6.8.5
309.	Elevation knob torque	3.7.2, 3.16.5	4.6.2, 4.6.8.6
310.	Azimuth knob torque	3.7.2, 3.16.5	4.6.2, 4.6.8.6
311.	Shock	3.7.3	4.6.3
312.	External tritium contamination (post temperature cycle, post shock)	3.1.2	4.6.5
313.	Illumination check (post temperature cycle, post shock)	3.1.3	Visual

NOTE: The tests for Characteristics 304 through 310 shall be conducted at temperatures of -50°F and +145°F. The tests for Characteristics 311 through 313 shall be conducted at a temperature between +60°F and +90°F.

4.4.3 Failure of sample.- Should any one item of a special sampling fail to meet the specified test requirements, acceptance of the represented inspection lot will be suspended by the Government until necessary corrections have been made by the contractor and resubmitted items have been approved. (see 4.3.3).

4.5 Inspection equipment.- Except as otherwise provided for by the contract, the contractor shall supply and maintain inspection equipment in accordance with the applicable requirements of MIL-I-45607.

4.5.1 Government furnished inspection equipment.- Where the contract provides for Government furnished test equipment, supply and maintenance of test equipment shall be in accordance with the applicable requirements specified in MIL-I-45607.

4.5.2 Contractor furnished inspection equipment.

4.5.2.1 Government design.- All inspection equipment specified by drawing number in specification or SQAP forming a part of the contract

Source: https://assist.dla.mil -- Downloaded: 2026-10-26T03:56Z
Check the source to verify that this is the current version before use.

shall be supplied by the contractor in accordance with technical data included in the technical data package list.

4.5.2.2 Contractor design.- In the absence of Government inspection equipment design the contractor shall design and supply inspection equipment compatible with the "Test Methods and Procedures" specified in 4.6 of this specification and with the component inspection procedures specified in "Examination" and "Test Facilities" requirements of MIL-F-13926. Since tolerance of test equipment is normally considered to be within 10% of the product tolerance for which it is intended, this inherent error in the test equipment design must be considered as part of the prescribed product tolerance limit. Thus, concept, construction, materials, dimensions and tolerances used in design of test equipment shall be so selected and controlled as to insure that the test equipment will reliably indicate acceptability of a product which does not exceed 90% of the prescribed tolerance limit, and permit positive rejection when non-conforming. Construction shall be such as to facilitate routine calibration of test equipment.

4.5.2.3 Inspection equipment design requirements.

4.5.2.3.1 Temperature testing.- The volume of the test chamber shall conform to the test facilities requirements of MIL-F-13926. The air temperature at the control sensor shall be in accordance with the specified requirements within plus or minus 2.5°F. The temperature gradient across the cross-sectional area occupied by the test item shall not exceed 0.5°F per foot in any direction. The heat source of the test chamber shall not be so located that radiant heat, if used, will not fall directly on the test item. Thermocouples or equivalent temperature sensors utilized to determine or control the specified chamber temperature shall be centrally located within the test chamber in the supply stream, or in the return airstream, whichever provides the specified test condition at the item under test and shall be baffled or otherwise protected against radiation effects.

4.5.2.3.2 Orientation.- The special testing equipment to position and test the elevation and azimuth accuracy of the mount shall conform to the following:

a. Elevation and Cross Leveling Fixture, F6523553
b. Azimuth Test Fixture, F7691596
c. Adapter, Final Inspection, Elevation, F11750708
d. Adapter, Final Inspection,Azimuth, F11750709
e. Alignment Gage, Final Inspection, Azimuth, F11750712
f. Inspection Aid, A8658940
g. Dioptometer, F7680631
h. Mounting Bracket for Dioptometer
i. Wall Target

Source: https://assist.dla.mil -- Downloaded: 2026-10-26T03:56Z
Check the source to verify that this is the current version before use.

4.5.2.3.2.1 Mounting bracket for dioptometer.- The mounting bracket shall consist of a sleeve and mandrel joined by a connecting arm. The geometric axis of the sleeve and mandrel shall lay in the same vertical plane within .001 inch. The geometric axes of the sleeve and mandrel shall also be horizontal and parallel to each other within .001 inch. The geometric axis of the sleeve shall be 1.50 + .25 inch above the geometric axis of the mandrel. The inside diameter of the sleeve shall be capable of accepting the Dioptometer, F7680631 (Outside diameter 1.18 - .01 inch). The sleeve shall secure and maintain the dioptometer in a horizontal position. The mandrel shall have a length of 1.000 + .250 inch and an outside diameter of .624 - .001 inch. Install the dioptometer in the sleeve of the bracket. Rotate the dioptometer until its vertical reticle line lays in the same vertical plane that contains the geometric axes of the sleeve and mandrel. Lock the dioptometer in this position.

4.5.2.3.2.2 Wall target.- A calibrated wall target shall be utilized in measuring the elevation accuracy requirements of the mount. The wall target shall contain a vertical and horizontal crossline pattern together with limiting displacement marks needed to determine line of sight deviations (azimuth shift) during elevation travels. This wall target shall be calibrated for a measured distance between 6.0 and 9.0 feet, as measured from the mechanical rotational axis (elevation) of the mount.

4.5.2.3.3 Elevation accuracy.- The testing equipment specified in 4.5.2.3.2a, 4.5.2.3.2c, 4.5.2.3.2f, 4.5.2.3.2g, 4.5.2.3.2h and 4.5.2.3.2i shall be utilized in performing the elevation travel and accuracy tests. The elevation and cross leveling fixture shall be positioned on a suitable stand and cross leveled in the horizontal plane by means of a certified 10 second level. The Adapter, F11750708 shall be affixed to the elevation and cross leveling fixture and cross leveled by means of the 10 second level so that the interfacing (female) dovetail meets the requirements of 3.8a and 3.8b.

4.5.2.3.4 Azimuth accuracy.- The testing equipment specified in 4.5.2.3.2b, 4.5.2.3.2d, 4.5.2.3.2e, 4.5.2.3.2g, 4.5.2.3.2h and 4.5.2.3.2i shall be utilized in performing the azimuth accuracy tests. The azimuth test fixture shall be positioned on a suitable stand and cross leveles in the horizontal plane by means of a certified 10 second level. Rotate the wheel assembly of the fixture to the zero azimuth setting. Lock the assembly in this position. Verify that the mechanical axis of the target projector collimator is aligned to the pivotal axis of the wheel assembly of the fixture. The height of the collimator shall be such that its optical axis is coincident to the axis of the sleeve of the mounting bracket when the mount, with mounting bracket and dioptometer affixed, is inserted in the Adapter, F11750709. With the Adapter, F11750709 mounted on the wheel assembly, the Alignment Gage, F11750712 together with the dioptometer shall align the adapter so that its interfacing (female) dovetail meets the requirement of 3.8a and 3.8b. The alignment gage, with dioptometer, shall also be utilized to establish the mounting dovetail surface of the mount parallel to the optical axis of the collimator.

Source: https://assist.dla.mil -- Downloaded: 2026-10-26T03:56Z
Check the source to verify that this is the current version before use.

4.5.2.3.5 Torque requirements.- The torque wrench type tester for testing knobs and clamping mechanism torque shall have a range of zero to twenty-five inch-pounds. The torque wrench type tester for testing rotation of the mandrel shall have a range of zero to fifty inch-pounds. The accuracy of the torque wrench type testers shall be five percent throughout its entire zone of excursion. The testing equipment shall be equipped with suitable adapters to accommodate testing of the appropriate assembly.

4.5.2.3.6 Fabrication, setup and calibration.- Design and fabrication of test equipment shall be such that it incorporates the features of rigidity, stability and ease of maintenance. Setup and calibration equipment necessary for certification of the testing equipment accuracy shall be furnished and will become a part of the testing equipment.

4.6 Test methods and procedures.

4.6.1 Storage temperature.- The testing equipment utilized in this test shall conform to the requirements of 4.5.2.3.1. The mount shall be placed in the test chamber and the temperature of the chamber reduced, over a period of 4 hours plus or minus 15 minutes, from room ambient temperature (60° to 90°F) to -50°F. The mount shall be maintained at this extreme temperature for a minimum of 4 hours. Upon completion of the 4 hour exposure the temperature of the test chamber shall be raised, over a period of 8 hours plus or minus 15 minutes, to +160°F. The mount shall remain at this temperature for a minimum of 4 hours. Upon completion of the 4 hour exposure the temperature of the chamber shall be reduced, over a period of 4 hours plus or minus 15 minutes, to room ambient temperature. The mount shall then be removed from the test chamber and subjected to the test in 4.6.5 and then a visual examination for illumination of the light sources. It shall meet the requirements of 3.1.2 and 3.1.3 respectively. The mount shall be subjected to a visual, tactile and audible examination. It shall show no evidence of physical damage. The mount shall then be subjected to the tests in 4.6.6 through 4.6.9 to determine overall conformance to the requirements of 3.7.1.

4.6.2 Operating temperature.- The testing equipment utilized in this test shall conform to the requirements of 4.5.2.3.1. The mount shall be placed in the test chamber and the temperature of the chamber reduced, over a period of 4 hours plus or minus 15 minutes, from room ambient temperature (60° to 90°F) to -50°F. The mount shall be maintained at this extreme temperature for a minimum of 4 hours. At the completion of the 4 hour exposure, and while at -50°F, the mount shall be subjected to the tests in 4.6.8.1 through 4.6.8.6. Upon completion of the low temperature operability tests the temperature of the chamber shall be raised, over a period of 8 hours plus or minus 15 minutes, to +145°F. The mount shall be maintained at this extreme temperature for a minimum of 4 hours. At the completion of the 4 hour exposure, and while at +145°F, the mount shall again be subjected to the tests in 4.6.8.1 through 4.6.8.6. Upon completion of the high temperature operability tests the temperature of the chamber shall be reduced, over a period of 4 hours plus or minus 15 minutes, to room ambient temperature. The mount shall then be removed from the test chamber and subjected to the test in 4.6.5 and then a visual examination for illumination of light sources.

Source: https://assist.dla.mil -- Downloaded: 2026-10-26T03:56Z
Check the source to verify that this is the current version before use.

It shall meet the requirements of 3.1.2 and 3.1.3 respectively subsequent to the temperature exposures of 3.7.2. The mount shall have conformed to the requirements of 3.7.2 at each temperature extreme.

4.6.3 Shock.- Conduct the shock test with the mount positioned on an adapter affixed to a shock test machine that conforms to the requirements of Drawing D7680617. The adapter shall be capable of holding and positioning the mount as specified in 3.7.3. The test shall be performed in accordance with Test Method 516.2, Procedure IV of MIL-STD-810, except that the number of shocks and applied directions shall be as specified in 3.7.3. The shock pulse shape, peak values and normal time durations for the positions indicated shall be as specified in 3.7.3. Condition the mount (scale and knobs) as specified prior to shock testing. Subsequent to the shock applications (18) the mount shall be removed from the shock test adapter and subjected to the tests in 4.6.5 and then a visual examination for illumination of the light sources. It shall meet the requirements of 3.1.2 and 3.1.3 respectively. The mount shall be subjected to a visual, tactile and audible examination. It shall show no evidence of physical damage. The mount shall then be subjected to the tests in 4.6.6 through 4.6.9 to determine overall conformance to the requirements of 3.7.3.

4.6.4 Vibration.- Conduct the vibration test with the mount positioned on an adapter affixed to a standard vibrating machine that is capable of providing the amplitude and frequency specified in 3.7.4. The adapter shall position the mount as specified in 3.7.4 during the vibration cycle. Unless otherwise specified the test equipment shall be in accordance with the "Test Facilities" requirements of MIL-F-13926. The applied motion shall be in a vertical plane for the time period specified. Upon completion of the vibration cycle the mount shall be removed from the vibration adapter and subjected to the test in 4.6.5 and then a visual examination for illumination of the light sources. It shall meet the requirements of 3.1.2 and 3.1.3 respectively. The mount shall be subjected to a visual, tactile and audible examination. There shall be no evidence of physical damage. The mount shall then be subjected to the tests in 4.6.6 through 4.6.9 to determine overall conformance to the requirements of 3.7.4.

4.6.5 External tritium contamination.- Contamination test shall be performed by wiping the mount with filter paper moistened in distilled water. The wiping shall be performed with moderate finger pressure. The damp filter paper shall be placed in a container with the proper portion of scintillation liquid. The scintillating system used to measure contamination must be calibrated and meet the minimum requirement of MIL-L-50778. The read out shall be within 10% of the known standard value when counted to a total of 2000 disintergrations. The actual test for determining contamination is performed by placing the container into the scintillation system. The measured contamination shall be within the limits specified in 3.1.2.

Source: https://assist.dla.mil -- Downloaded: 2026-10-26T03:56Z
Check the source to verify that this is the current version before use.

4.6.6 Orientation, elevation accuracy.- The testing equipment utilized for these tests shall conform to the requirements of 4.5.2.3.3. Position the mount on the elevation adapter. Orient the mount as specified in 3.8d through 3.8g. Adjust the dioptometer eyepiece for best focus of the dioptometer reticle. Insert the mandrel portion of the mounting bracket, with dioptometer, into the telescope clamping mechanism of the mount. The eyepiece of the dioptometer shall be on the side of the mount that contains the elevation knob. Rotate the mounting bracket, in the clamping mechanism, until the dioptometer vertical reticle line is plumb. Lock the clamp mechanism. Position the wall target at its calibrated distance. Focus the dioptometer on the wall target by means of the objective assembly. Elevate the target until the intersecting point of the horizontal and vertical target line is coincident to the intersecting point of the dioptometer horizontal and vertical reticle lines. Plumb the vertical target line independent of the dioptometer vertical reticle line.

NOTE: The condition of 3.8d shall be established with backlash removed from the elevation mechanism for rotation in depression.

4.6.6.1 Elevation travel.- With backlash removed for direction of travel, and while directing a line of sight through the center of the dioptometer reticle pattern to the vertical target line, depress the line of sight of the mount 200 mils by means of its elevation knob. Return the line of sight, through the dioptometer, to the horizontal plane by elevating the elevation and cross leveling fixture. Repeat the above steps until the mount is at maximum depression. Position the Inspection Aid, 8659940 on the quadrant seats provided on the elevation adapter and verify the mil value indicated on the elevation scale and micrometer knob of the mount, by measuring the angle established on the elevation adapter. Return the mount and elevation and cross leveling fixture to zero mil elevation. By means of the elevation knob, elevate the line of sight of the mount to its maximum elevation position. Again, by means of the Inspection Aid 8658940, verify the mil value indicated on the elevation scale and micrometer knob of the mount. The mount shall conform to the requirements of 3.9.

4.6.6.2 Plumb travel.- Perform this test concurrent with the test in 4.6.6.1. During the excursions in depression and elevation the vertical reticle line of the dioptometer, at the center of the reticle pattern, shall track the vertical target line within the tolerance specified in 3.10, as measured on the target. Caution shall be exercised to assure that this test is conducted with backlash removed for direction of travel.

4.6.6.3 Lateral displacement.- Perform this test in sequence to the test in 4.6.6.1 and 4.6.6.2. With backlash removed for direction of travel, and the dioptometer reticle pattern superimposed on the vertical target line as specified in 4.6.7.1, track the vertical target line for approximately 50 mils by means of the elevation knob of the mount. Then, with backlash included, reverse direction of the elevation knob for approximately 50 mils. Any lateral displacement of the dioptometer vertical reticle line, as a result of this operation, shall not exceed the tolerance specified in 3.11 as measured on the target. Tests for lateral displacement shall be performed

95

Source: https://assist.dla.mil -- Downloaded: 2026-10-26T03:56Z
Check the source to verify that this is the current version before use.

within 50 mils of each extremity of the elevation scale. A third test point for lateral displacement shall be performed at approximately mid-range of the elevation scale. All points of tests for lateral displacement shall be randomly selected within the above specified areas.

4.6.6.4 Elevation angular accuracy.- Perform this test at a minimum of four randomly selected elevation settings in depression (mil value positions above 800 mils) and one elevation setting below the 800 mil scale setting. With backlash removed for fixture rotation in elevation, elevate the adapter to zero mil in elevation. The zero mil elevation position of the elevation adapter shall be verified with the Inspection Aid, A8658940. With backlash removed for rotation of a line of sight in depression verify that the dioptometer reticle lines are superimposed on the target lines, and that the mount conforms to the orientation requirements of 3.8d through 3.8g. Depress the line of sight of the mount, by means of its elevation knob, to the first randomly selected elevation setting. Then by means of the elevation mechanism of the test fixture, return the line of sight of the mount to the horizontal plane until the dioptometer reticle lines are again superimposed on the target lines without overtravel. Utilizing the inspection aid, measure the elevation angle on the adapter. The measured angle recorded on the inspection aid must equal the angle indicated on the scale and micrometer of the mount within the tolerance specified in 3.14.1. Utilizing the foregoing method proceed to inspect the elevation accuracy of the mount at the three remaining selected elevation settings. The one elevation setting below 800 mils shall be inspected from a zero mil setting of the fixture and a 800 mil setting of the mount. Record the elevation error for each elevation setting tested for use in the elevation backlash test. Caution must be exercised throughout all elevation accuracy measurements to exclude backlash from the mount and test fixture for their appropriate directions of travel. The elevation accuracy of the mount shall conform to the requirements of 3.14.1.

4.6.6.5 Elevation backlash.- Perform this test in sequence to the test in 4.6.6.4. At each of the selected elevation setting record the elevation (mil) value when the reticle lines are in coincident with the target lines. With backlash excluded for direction of travel override the target line (horizontal) by approximately 50 mils utilizing the elevation knob of the mount. Then with backlash included reverse direction of the elevation knob and reestablish coincidence of the reticle and target lines (horizontal), without overtravel. Again record the elevation mil value as read on the scale and micrometer knob. The difference between the first and second recorded mil value is the amount of backlash. At the selected elevation setting the amount of backlash for that elevation position, added to the elevation accuracy error recorded for that elevation position shall not exceed the tolerance specified in 3.15.1. Perform the elevation backlash test at each of the selected elevation settings tested in 4.6.6.4.

4.6.7 Orientation, azimuth accuracy.- The testing equipment utilized for these tests shall conform to the requirements of 4.5.2.3.4. Position the mount on the azimuth adapter. Adjust and position the dioptometer and mounting bracket on the mount as specified in 4.6.6. Orient the mount as

Source: https://assist.dla.mil -- Downloaded: 2026-10-26T03:56Z
Check the source to verify that this is the current version before use.

specified in 3.8d through 3.8g. Verify that the vertical and horizontal reticle lines of the dioptometer and collimator target lines are superimposed. Then ascertain that the optical axis of the dioptometer is horizontal by rotating the wheel assembly of the fixture to 1600 mils, then counter rotating the mount (by azimuth knob) to return the dioptometer vertical reticle line coincident to the collimator vertical target line. Correct any elevation error, as evidenced by the lack of coincidence of the horizontal reticle and collimator target lines, by adjusting the clamping mechanism for elevation of the line of sight. Then rotate the wheel assembly of the fixture to 4800 mils and repeat the above procedure to level the dioptometer. Repeat the entire procedure until the dioptometer is horizontal at 1600 mils, zero mils and 4800 mils, without interim adjustments. Return the fixture to the zero mil azimuth setting and the dioptometer reticle coincident to the target collimator vertical and horizontal lines.

4.6.7.1 Horizontal travel.- Verify that the intersecting point of the dioptometer reticle lines is coincident with the horizontal target line. With backlash removed for direction of travel, rotate the line of sight of the mount through 6400 mils, in precise 800 mil increments. At each 800 mil incremental position of the mount, counter-rotate the wheel assembly of the fixture 800 mils to reestablish coincidence of the dioptometer vertical reticle line and vertical target line. At each incremental position note any lack of coincidence of the horizontal reticle line and horizontal target line. Any vertical displacement of the line of sight through the intersecting point of the dioptometer reticle lines, with reference to the horizontal target line, shall not exceed the tolerance specified in 3.13 as measured on the target collimator.

NOTE: Care shall be exercised to assure that backlash is excluded in all horizontal travel measurements. The 800 mil incremental positions of the mount and test fixture shall be accomplished without overtravel.

4.6.7.2 Vertical displacement.- This test may be conducted concurrent with the test in 4.6.7.1. At each 800 mil incremental position reverse the azimuth motion of the mount by means of the azimuth knob. As a result of this reversing operation the line of sight, through the intersection of the dioptometer reticle lines, shall not displace vertically in excess of the tolerance specified in 3.12, as measured on the target collimator.

4.6.7.3 Azimuth angular accuracy.- Perform this test at 800 mil incremental positions (8) as the line of sight of the mount is rotated 6400 mils in azimuth. Verify that the fixture wheel assembly is positioned at zero mil azimuth with backlash removed for appropriate movement in azimuth. With backlash removed for rotation of the line of sight of the mount to 800 mils verify that the dioptometer vertical reticle line is superimposed on the vertical target line, and that the mount conforms to the requirements of 3.8d through 3.8g. Rotate the line of sight of the mount to 800 mils without overtravel. Counter rotate the fixture wheel assembly and again obtain coincidence of the dioptometer vertical reticle line and vertical target line without overtravel. Any difference between the mil value

Source: https://assist.dla.mil -- Downloaded: 2026-10-26T03:56Z
Check the source to verify that this is the current version before use.

recorded on the mount and the scale of the fixture is azimuth accuracy error. Record any azimuth accuracy error for use in the azimuth backlash test. Utilizing the foregoing method proceed to inspect the azimuth accuracy of the mount at the 7 remaining incremental positions. Record the azimuth error for each of the 8 incremental positions tested. Caution must be exercised throughout all azimuth accuracy measurements to exclude backlash from the mount and test fixture for their appropriate direction of travel. The azimuth accuracy of the mount shall conform to the requirements of 3.14.2.

4.6.7.4 Azimuth backlash.- Perform this test in sequence to the test in 4.6.7.3, at each of the 8 incremental positions. Subsequent to the azimuth angular accuracy measurements rotate the mount, by means of the azimuth knob, to an azimuth setting of approximately 850 mils. Then with backlash included, reverse direction of the knob and again obtain coincidence of the dioptometer vertical reticle line and vertical target line without overtravel. Record the azimuth mil value as read on the azimuth knob. Any azimuth change from the original 800 mil setting, as read on the azimuth knob scale, is the amount of backlash. At the selected azimuth position the amount of backlash for that position, added to the azimuth accuracy error recorded for that position shall not exceed the tolerance specified in 3.15.2.

4.6.8 Operability.- The following test shall be conducted with the mount positioned on a suitable stand that as a mating dovetail that maintains the mount as specified in 3.8a and 3.8b. A mandrel that meets the physical dimensions specified in 4.5.2.3.2.1 and torque testing equipment conforming to 4.5.2.3.5 shall also be utilized.

4.6.8.1 Telescope clamping mechanism, unlocked position.- When subjected to a tactile examination the function of the locking lever shall meet the requirements of 3.16.1.1. When the mandrel is inserted into the telescope holder it shall rotate freely for 360 degrees.

4.6.8.2 Telescope clamping mechanism, locked position.- Insert the mandrel into the telescope holder. Apply a force of 10 inch-pounds to the locking lever. Apply the specified torques to the mandrel and telescope holder, in turn. The telescope clamping mechanism shall meet the requirements of 3.16.1.2.

4.6.8.3 Mount locking latch.- Install and remove the mount from the dovetail on the stand a minimum of 3 times (3 installations). The mount shall meet the requirements of 3.16.2, at the temperature specified.

4.6.8.4 Locking knobs.- Rotate the azimuth and elevation locking knobs to a locked position with the appropriate torque wrench and adapter. The azimuth and elevation mechanisms shall meet the requirements of 3.16.3 when the specified torque is applied to the applicable knob, at the temperature indicated.

Source: https://assist.dla.mil -- Downloaded: 2026-10-26T03:56Z
Check the source to verify that this is the current version before use.

4.6.8.5 Azimuth slip scales.- When subjected to a tactile examination for operability, the micrometer knob slip scales (azimuth and elevation) and azimuth scale shall meet the requirements of 3.16.4, at the temperature indicated.

4.6.8.6 Torque, azimuth and elevation knobs.- Utilizing the appropriate torque wrench with adapter, the running torque required to rotate the knobs throughout their range of rotation shall be within the values specified in 3.16.5, for the temperature indicated.

4.6.9 Illumination.- The illumination test shall be performed under simulated conditions specified in 3.17. Place the mount in a dark room having a controlled light source. The observer and mount must become dark adapted. Perform the test by observing with the unaided eye, from a distance of 18 to 20 inches, and determining that all scale markings, indicies, level vial bubbles and markings are clearly discernible in the dark room (controlled light source "OFF"). Gradually raise the level of illumination to 5 foot-candles. The scale markings, indicies, level vial bubbles and markings shall meet the requirements of 3.17.

4.6.10 Workmanship.- The mount shall be subjected to a visual and tactile examination for evidence of faulty workmanship. The mount shall conform to the requirements of 3.6.

4.6.11 Reliability demonstration.- The Probability of Success of 96% (para 3.18) shall be demonstrated at a 90% confidence level. For this test, successful completion of all the optical/mechanical checks is considered a mission. To demonstrate the reliability requirement, the number of missions required with the associated failure is depicted in Table V. The reliability demonstration test shall be performed as part of First Article Testing. After each individual environmental exposure of para 3.7 the optical/mechanical check shall be performed. At the conclusion of the environmental exposure, each additional mission required by Table V to prove the reliability requirement shall consist of exposure to the shock requirement of para 3.7.3 followed by successful completion of the optical/mechanical checks. The total number of missions with associated failed mission number shall be compared to Table V for acceptance.

4.6.11.1 Reliability failure.- A reliability failure is considered to be failure to meet all the optical/mechanical checks of para 3.8.

TABLE #V

Total Missions	Failed Missions	
	Accept	Reject
57	0	5
96	1	5
132	2	5
166	3	5
198	4	5

Source: https://assist.dla.mil -- Downloaded: 2026-10-26T03:56Z
Check the source to verify that this is the current version before use.

5. PREPARATION FOR DELIVERY

5.1 Packaging, packing and marking.- Packaging, packing and marking of the Mount, Telescope 11741796 shall be in accordance with Packaging Data Sheet 11741796. The level of protection shall be as specified in the procurement document.

6. NOTES

6.1 Intended use.- The Mount Telescope, 11741796 is a component of Sightunit: XM64E2. The Mount introduces measured azimuth and elevation readings to Elbow Telescope: 11741796. The Sightunit is primarily used with the Lightweight Company Mortar, 60MM, XM224.

6.2 Ordering data.- Procurement documents should specify the following:

(a) Title, number and date of this specification.
(b) Selection of applicable levels of preservation, packaging and packing.
(c) Applicable stock number.
(d) Applicable Packaging Data Sheet Number (See 5.1).
(e) Provisions for First Article Testing.

Custodian

Army - MU

Preparing activity

Army - MU

Project No. 1240-A706

Source: https://assist.dla.mil -- Downloaded: 2026-10-26T03:56Z
Check the source to verify that this is the current version before use.

Source: https://assist.dla.mil -- Downloaded: 2026-10-26T03:56Z
Check the source to verify that this is the current version before use.

STANDARDIZATION DOCUMENT IMPROVEMENT PROPOSAL

OMB Approval
No. 22-R255

INSTRUCTIONS: The purpose of this form is to solicit beneficial comments which will help achieve procurement of suitable products at reasonable cost and minimum delay, or will otherwise enhance use of the document. DoD contractors, government activities, or manufacturers/ vendors who are prospective suppliers of the product are invited to submit comments to the government. Fold on lines on reverse side, staple in corner, and send to preparing activity. Comments submitted on this form do not constitute or imply authorization to waive any portion of the referenced document(s) or to amend contractual requirements. Attach any pertinent data which may be of use in improving this document. If there are additional papers, attach to form and place both in an envelope addressed to preparing activity.

DOCUMENT IDENTIFIER AND TITLE

MIL-M-48575(MU) Mount, Telescope: 11741796

NAME OF ORGANIZATION AND ADDRESS

CONTRACT NUMBER

MATERIAL PROCURED UNDER A

☐ DIRECT GOVERNMENT CONTRACT ☐ SUBCONTRACT

1. HAS ANY PART OF THE DOCUMENT CREATED PROBLEMS OR REQUIRED INTERPRETATION IN PROCUREMENT USE?

A. GIVE PARAGRAPH NUMBER AND WORDING.

B. RECOMMENDATIONS FOR CORRECTING THE DEFICIENCIES

2. COMMENTS ON ANY DOCUMENT REQUIREMENT CONSIDERED TOO RIGID

3. IS THE DOCUMENT RESTRICTIVE?

☐ YES ☐ NO *(If "Yes", in what way?)*

4. REMARKS

SUBMITTED BY *(Printed or typed name and address - Optional)*

TELEPHONE NO.

DATE

DD FORM 1 JAN 72 1426 REPLACES EDITION OF 1 JAN 66 WHICH MAY BE USED

S/S

Source: https://assist.dla.mil -- Downloaded: 2026-10-26T03:56Z
Check the source to verify that this is the current version before use.

Source: https://assist.dla.mil -- Downloaded: 2026-10-26T03:56Z
Check the source to verify that this is the current version before use.